

Governed Multi-Agent GraphRAG for Banking Regulatory Intelligence

Phase 4 — AWS V2

From GraphRAG Proof of Concept to a Production-Oriented,
Governed Multi-Agent Architecture on AWS with Databricks

LangGraph | Microsoft GraphRAG | MCP | OpenAI GPT-4.1 | DeepSeek V4 Flash | AWS ECS | Databricks | LangSmith

Jingru Chen • Phase 4 Complete — September 28, 2026

From GraphRAG Foundation to AWS ECS + Databricks Delivered Architecture

2

The portfolio evolves from evidence-grounded GraphRAG retrieval into a governed, deployable multi-agent AI runtime.

PHASE 1 — FOUNDATION

Microsoft GraphRAG PoC

- 23-document banking / regulatory corpus
- Entity, relationship & community indexing
- Basic / Local / Global Search
- Evidence validation & citation checks
- Runtime guardrails
- Public Streamlit demonstration

PHASE 4 (AWS V2) — DELIVERED

Governed Multi-Agent GraphRAG + MCP

- LangGraph planning / research / validation workflow
- Hybrid retrieval: Existing + **MCP** + Global Search, including current regulatory evidence
- Deterministic guardrails + LLM judge
- Retry / recovery / escalation / human review
- OpenAI + DeepSeek multi-provider runtime
- Docker → ECR → ECS public **AWS deployment**
- **Databricks** Delta query-history persistence
- LangSmith + CloudWatch observability

PHASE 5 (AWS V3) — Next

Harness Engineering & Production Optimization

- **Harness Engineering** for systematic E2E evaluation
- Regression test suites across retrieval, guardrail, judge & recovery
- Automated provider / routing / failure-injection testing
- Global Search optimization for quality, latency, cost & long-running execution
- Expanded MCP / enterprise data-source integration
- Task-level intelligent model routing
- Production scale, latency & cost optimization

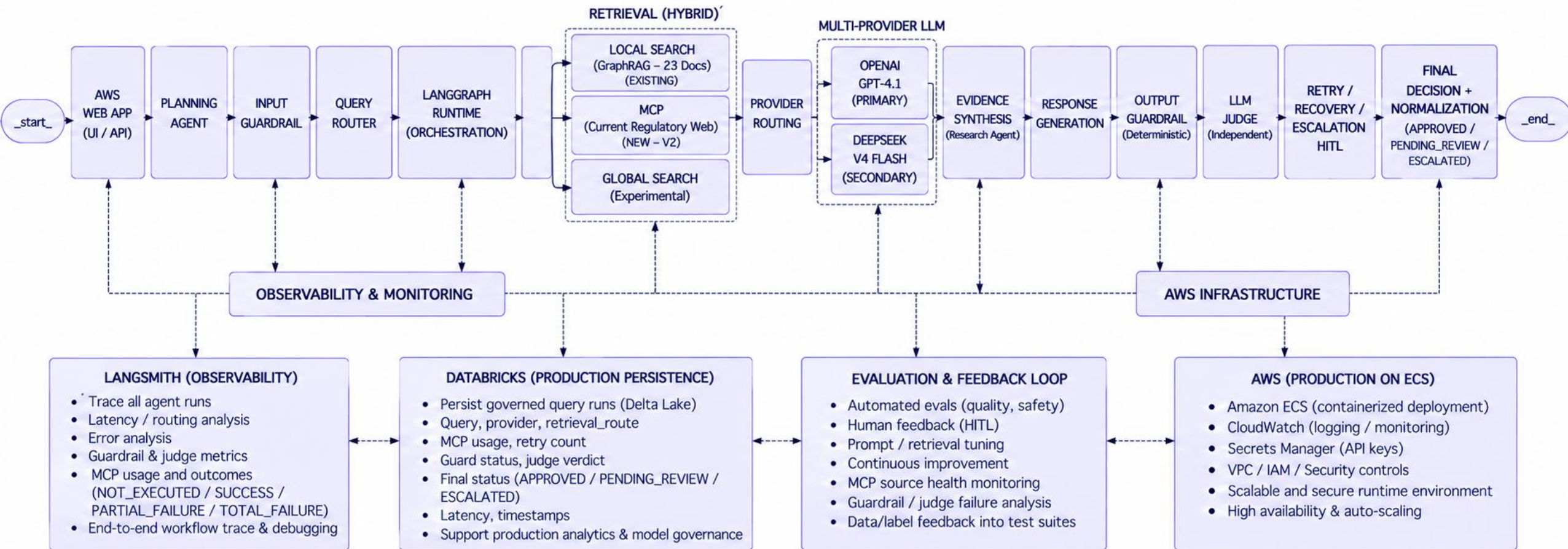
[Phase 1 — GraphRAG PoC Demo](#)

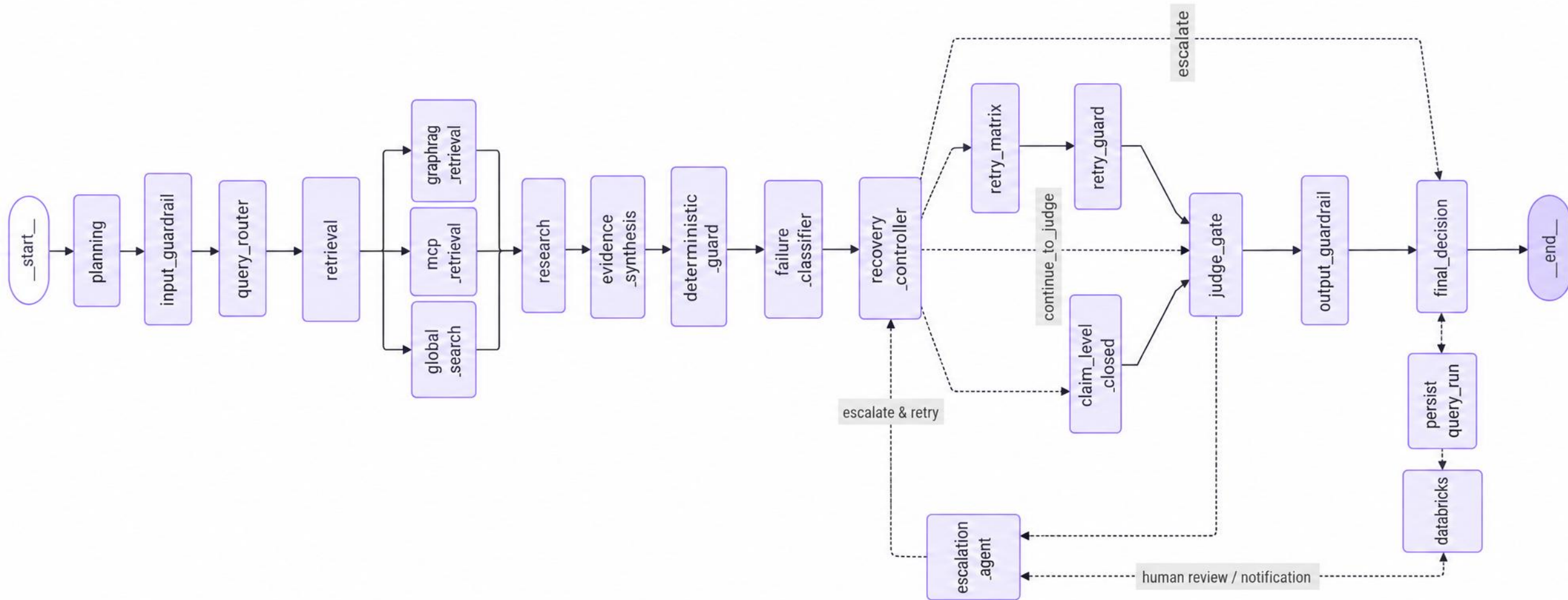
[Phase 4 — AWS V2 Live Demo](#)

Engineering story: identify failure modes → design governed orchestration → validate end-to-end → deploy on AWS → observe with LangSmith → persist runtime outcomes in Databricks.

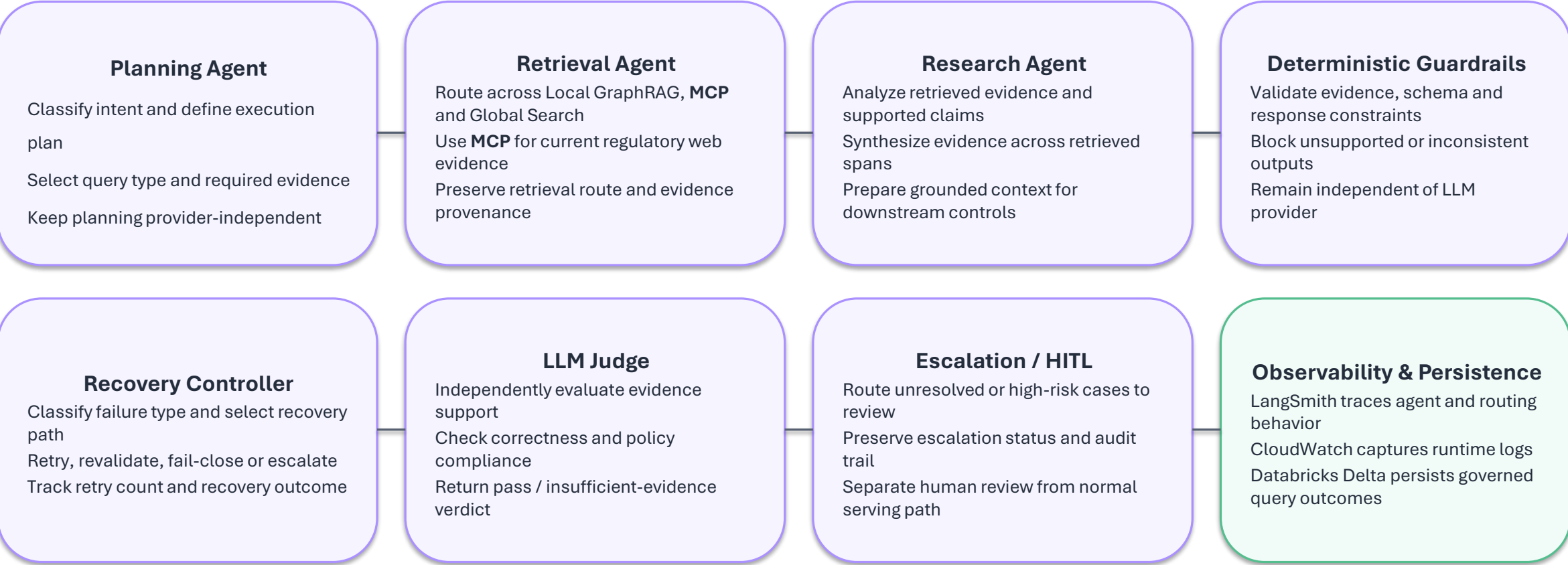
End-to-End Architecture for Governed Multi-Agent GraphRAG with Multi-Provider LLM

Enhanced Version (V2.0) – Deployed on AWS ECS and Databricks





Separation of responsibilities across planning, retrieval, research, governance, recovery, judging, escalation and runtime persistence



Design principle: separate planning, retrieval, research, deterministic governance, recovery and judging — then observe with LangSmith and persist governed runtime outcomes in Databricks.

Defense-in-depth controls convert retrieval, evidence, provider, and MCP failures into explicit, recoverable, and auditable runtime decisions.

1 PREVENT

Plan & Input Guardrails

Validate request and execution plan

Bound unsupported behavior

Enforce governed routing constraints

Preserve provider-independent controls

2 DETECT

Deterministic Guard + Failure Classifier

Evidence sufficiency checks

Citation / provenance validation

Detect missing or unsupported evidence

Classify retrieval / MCP / provider failures

3 RECOVER

Recovery Controller

Select retry / revalidation strategy

Apply retry matrix + retry guard

Re-run controlled retrieval when appropriate

Claim-level fail-closed when recovery fails

4 JUDGE

Independent LLM Judge

Separate evaluation from answer generation

Validate evidence-supported output

Apply judge contract guardrail

Pass, reject, or route unresolved cases

5 ESCALATE

Governed Escalation / HITL

PENDING_REVIEW is a valid governed outcome

Escalate unresolved or high-risk cases

Preserve review / escalation status

Never convert missing evidence into unsupported answers

Governance objective: make failures explicit, recoverable, auditable, and safe — rather than maximizing answer rate at the expense of evidence quality.

Layered Guardrail System — Defense in Depth

7

Guardrails operate across planning, hybrid retrieval, evidence, output contracts, recovery, and independent review—not as a single post-processing check.

L1 REQUEST & PLAN

Prevent invalid or unsafe execution

- Validate request and plan structure
- Bound unsupported / out-of-scope behavior
- Preserve selected provider and governed routing
- Reject malformed planning outputs

L2 RETRIEVAL & SCOPE

Constrain what evidence may enter

- Route Local GraphRAG / MCP / Global Search
- Bank / document / period scope checks
- Canonical source-ID + alias resolution
- Trusted-source and span-catalog boundaries

L3 EVIDENCE & PROVENANCE

Prove every factual claim

- Evidence sufficiency / missing-evidence checks
- Citation-to-source provenance validation
- Exact quote / span validation
- Value, unit, period & scope field evidence

L4 OUTPUT CONTRACT

Validate the governed research matrix

- Schema / field contract validation
- Supported ⇒ evidence for every required field
- Bound evidence spans per governed row
- No validated governed output ⇒ no public answer

L5 FAILURE & RECOVERY

Fail closed, then recover selectively

- Classify retrieval / MCP / provider failures
- Retry matrix + retry guard
- Revalidate evidence after controlled recovery
- Claim-level fail-closed / escalation

L6 JUDGE & HUMAN REVIEW

Independent final governance

- Independent LLM Judge review
- Judge-contract guardrail
- APPROVED / PENDING_REVIEW / ESCALATED outcomes
- Escalate unresolved risk to human review

Control principle: validate early → govern retrieval → ground every claim → recover narrowly → revalidate → judge independently → escalate when uncertainty remains.

V2 was frozen after AWS runtime, governed query execution, LangSmith tracing, and Databricks persistence were validated end-to-end.

DEPLOYMENT PATH

Application + governed LangGraph runtime
↓ Docker container → Amazon ECR
↓ Amazon ECS public service
Health endpoint + service rollout validated

AWS V2 FREEZE EVIDENCE

ECS deployment: COMPLETED / stable
Desired: 1 Running: 1 Pending: 0
Health endpoint: PASS
Governed /query execution: PASS
Final V2 runtime: CLOSED PASS

CROSS-PLATFORM E2E EVIDENCE

AWS governed query: PASS
LangSmith trace: PASS
Databricks query_runs persistence: PASS
MCP retrieval + outcomes observable
OpenAI / DeepSeek runtime evidence captured

MCP GOVERNANCE — NEXT HARDENING AREA

MCP retrieval can return current regulatory evidence successfully.
Dynamic web evidence still needs MCP-aware provenance, schema normalization, evidence-sufficiency and judge rules.
Current controls may reject successful MCP retrieval results → benchmark case for Harness Engineering.

GLOBAL SEARCH — VALIDATED, STILL EXPERIMENTAL

Global Search backend is validated, but long-running synchronous public-AWS execution is not production-ready.
Next: async job execution → job_id → background processing → polling / result retrieval.
Also tune quality, latency and cost before production promotion.

AWS ECS: PASS

LANGSMITH: PASS

DATABRICKS: PASS

**MCP RETRIEVAL:
VALIDATED**

**MCP GOVERNANCE:
HARDENING**

GLOBAL SEARCH: EXPERIMENTAL

Freeze principle: deploy → validate governed behavior → verify observability → verify persistence → document remaining hardening gaps → freeze proven layers.

LangSmith

Application

All applications

Search

Home

Tracing

Engine

Monitoring

Datasets & Experiments

Evaluators

Annotation Queues

Custom Apps

Prompts

Playground

Studio

Context Hub

Deployments

Sandboxes

LLM Gateway

Settings

Personal

chen.jingru@gmail.com

Personal / Tracing / GraphRAG-Phase4-AWS-V2.0

GraphRAG-Phase4-AWS-V2.0

Tracing

Evaluators

Automations

Insights

Engine

Retention 14d

Dashboard

New

in root run

Search...

Shortcuts

Last 1 day

Threads

Traces

Runs

Views

Stats · 15 traces

Status

success

error

interrupted

pending

Run Name

LangGraph

Run Type

chain

Metadata

LANGSMITH_PROJE...

LANGSMITH_TRACI...

Is_integration

Is_run_depth

		Name	Input	Output	Error	Start Time	Latency	Dataset	Annotation ...
☐	☒	LangGraph	openai	REGULATORY		9/27/2026, 12:33:55 AM	8.40s	☐	☐
☐	☒	LangGraph	openai	PENDING_REVIEW		9/27/2026, 12:33:26 AM	9.87s	☐	☐
☐	☒	LangGraph	openai	REGULATORY		9/27/2026, 12:32:59 AM	9.04s	☐	☐
☐	☒	LangGraph	openai	REGULATORY		9/27/2026, 12:29:20 AM	8.96s	☐	☐
☐	☒	LangGraph	deepseek	REGULATORY		9/27/2026, 12:28:45 AM	7.93s	☐	☐
☐	☒	LangGraph	openai	REGULATORY		9/27/2026, 12:23:42 AM	8.84s	☐	☐
☐	☒	LangGraph	deepseek	REGULATORY		9/27/2026, 12:22:51 AM	9.39s	☐	☐
☐	☒	LangGraph	openai	PENDING_REVIEW		9/27/2026, 12:20:33 AM	9.24s	☐	☐
☐	☒	LangGraph	deepseek	PENDING_REVIEW		9/27/2026, 12:20:10 AM	10.16s	☐	☐
☐	☒	LangGraph	openai	METRIC		9/27/2026, 12:14:54 AM	15.66s	☐	☐
☐	☒	LangGraph	openai	DESCRIPTIVE		9/27/2026, 12:08:58 AM	17.00s	☐	☐
☐	☒	LangGraph	openai	DESCRIPTIVE		9/26/2026, 11:34:00 PM	18.73s	☐	☐
☐	☒	LangGraph	deepseek	REGULATORY		9/26/2026, 9:09:55 PM	7.28s	☐	☐

LangSmith

Application

All applications

Search

Home

Tracing31

EnginePlus

Monitoring1

Datasets & Experiments3

Evaluators

Annotation Queues2

Custom Apps

Prompts

Playground

Studio

Context Hub

DeploymentsPlus

Sandboxes

LLM GatewayPlus

Keyboard Shortcuts

Settings

Personal

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Personal / Tracing / GraphRAG-Phase4-AWS-V2.0 / LangGraph

LangGraph

ID

Trace

Waterfall

Filter runs...

0s1s2s3s4s5s6s7s8s

LangGraph 9.04s

runtime_bootstrap 2.83s

planning_agent 1.38s

plan_guardrail 0.00s

route_plan_guardrail 0.00s

retrieval 2.34s

span_catalog 0.00s

evidence_context_enricher 0.00s

research_agent 1.58s

deterministic_guard 0.00s

failure_classifier 0.00s

recovery_controller 0.00s

route_recovery 0.00s

claim_level_fail_closed 0.00s

deterministic_revalidation 0.00s

route_revalidation 0.00s

judge_gate 0.00s

route_judge_gate 0.00s

llm_judge 0.86s

LangGraph

ID

Feedback

Input

Output

Attributes

Input

Output

Fields

final_status PENDING_REVIEW

guard_status BLOCK

intent_type REGULATORY

judge_contract_status PASS

judge_verdict insufficient_evidence

mcp_outcome SUCCESS_WITH_RESULTS

provider openai

query What is the FDIC's current guidance on model risk management?

recovery_action CLAIM_LEVEL_FAIL_CLOSED

research_matrix_raw { "rows": [{ "guidance": "FDIC Model Risk Management Guidanc...

retrieval_capability_reason EXPLICIT_CURRENT_REGULATORY_INTENT

retrieval_capability_route MCP

revalidation_status PASS

schema_type GUIDANCE

bootstrap_seconds 2.830849662999981

evidence_insufficiency []

AWS V2 E2E Runtime Evidence — LangSmith Observability & Databricks Persistence

11

← ↻ 🔒 https://dbc-a354c4a3-c610.cloud.databricks.com/editor/queries/3568729127001441?o=7474644240313583&contextId=sql-editor

databricks
Free Edition

Search data, notebooks, recents, and more... CTRL + P

Verify identity workspace

+ New

Home Learn Workspace Recents Catalog Jobs & Pipelines Compute Discover Marketplace

SQL

SQL Editor

Queries Dashboards Genie Agents Alerts Query History SQL Warehouses

Data Engineering

Runs Data Ingestion Visual Data Prep

AI/ML

Playground

SQL Editor

GraphRAG_Phase3_StepV2_13_DATABRICKS_P... x +

Run all (1000) 12:29 AM (1s) workspace default Last edit was 12 hours ago

Edit Serverless Starte... 2XS Schedule Save* Comments Share

```
117 /
118 SELECT
119     run_id,
120     query,
121     provider,
122     retrieval_route,
123     mcp_used,
124     retry_count,
125     guard_status,
126     judge_verdict,
127     final_status,
128     latency_ms,
129     created_at
130 FROM workspace.multi_agent_graphrag.query_runs
131 ORDER BY created_at DESC
132 LIMIT 10;
```

Add parameter

Table +

	query	provider	retrieval_route	mcp_used	retry_count	guard_status	judge_verdict
1	What is the FDIC's current guidance on model risk management?	openai	MCP	true	0	PASS	insufficient_evidence
2	What is the FDIC's current guidance on model risk management?	deepseek	HYBRID	true	1	BLOCK	null
3	What is the OCC's current guidance on model risk management?	openai	MCP	true	0	PASS	insufficient_evidence
4	What is the OCC's current guidance on model risk management?	deepseek	HYBRID	true	1	BLOCK	null
5	What is the Federal Reserve's current guidance on model risk management?	openai	HYBRID	true	0	BLOCK	insufficient_evidence
6	What is the Federal Reserve's current guidance on model risk management?	deepseek	HYBRID	true	1	PASS	insufficient_evidence
7	What was Wells Fargo's provision for credit losses for loans for the quarter ended June 30, 2026?	openai	EXISTING	false	0	PASS	pass
8	What are Citigroup's major credit risk exposures?	openai	EXISTING	false	1	BLOCK	null
9	STEP V2-13F-09F-03B persistence idempotency smoke test	deepseek	HYBRID	true	1	TEST_ONLY	null

9 rows 0.810s runtime See performance (1) ✓ Checked for improvements

Refreshed at 9/27/26, 12:29 AM (11h ago)